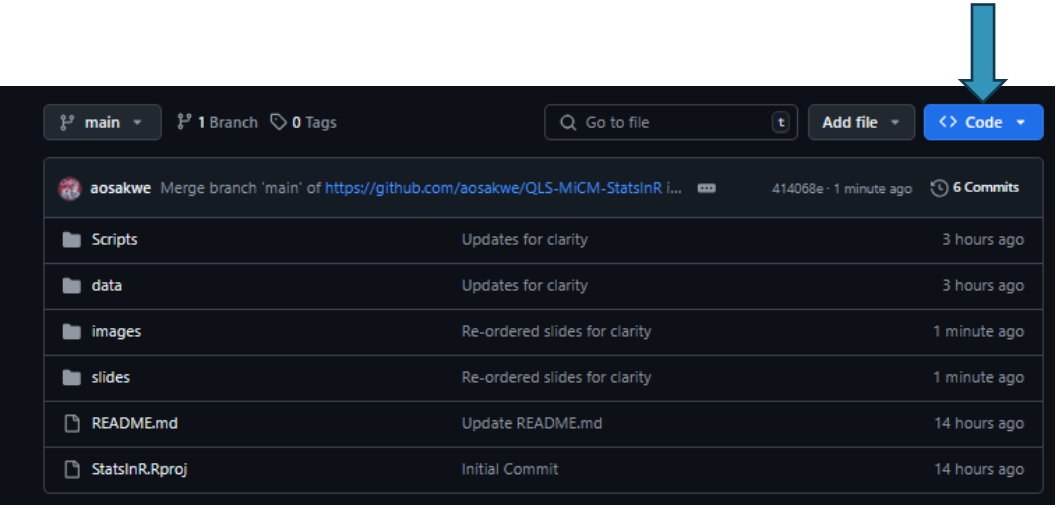
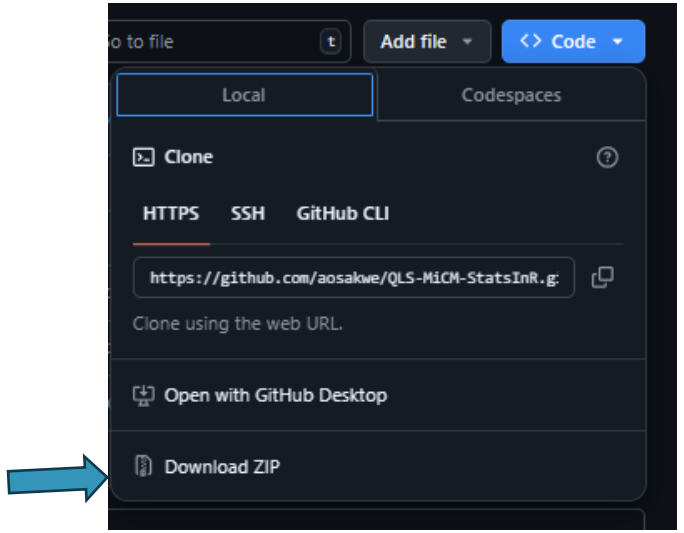
# Workshop materials

https://github.com/QLS-MiCM/StatsInR

use ‘git clone’ command OR



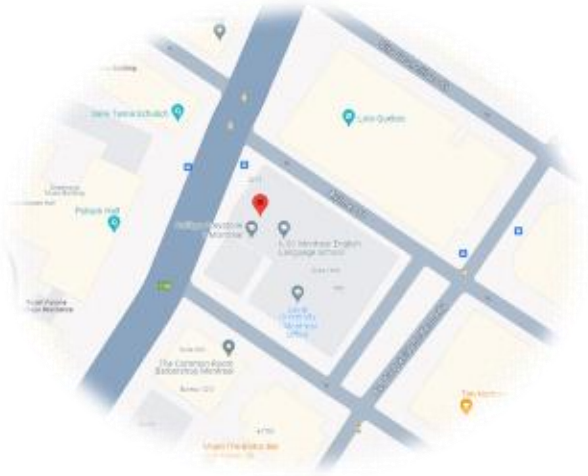
# Statistical Analysis in R

Workshop Lead: Sameena Karsan
October 19, 2026

**Slides are adapted from slides created by Adrien Osakwe for this workshop in May.**

MiCM McGill initiative in Computational Medicine

QLS-MiCM mission statement: deliver quality workshops designed to help biomedical researchers develop the skills they need to succeed.



Location: 550 Sherbrooke Street, Montreal, Quebec

Scan the QR code to sign up for our **mailing list**

Contact: workshop-micm@mcgill.ca

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# Workshop Series

| Workshop | Date | Location | Registration |
|---|---|---|---|
| **How to Think in Code** | September 22, 1 - 3 PM | Sherbrooke 550 W, Rm 136 | Closed |
| **Intro to Git & GitHub** | September 23, 1 - 3 PM | Sherbrooke 550 W, Rm 136 | Closed |
| **Intro to Unix** | September 25, 1 - 3 PM | Sherbrooke 550 W, Rm 136 | Closed |
| **Intro to DRAC/Compute Canada** | September 28, 10 AM - 12 PM | Education Building, 3700 McTavish, Rm 133 | Closed |
| **Intro to R (Part 1)** | September 30, 1 - 3 PM | Sherbrooke 550 W, Rm 136 | Closed |
| **Intro to Python (Part 1)** | October 2, 1 - 3 PM | Sherbrooke 550 W, Rm 189 | Closed |
| **Stats in R (Part 2)** | October 19, 11 AM - 1 PM | Education Building, 3700 McTavish, Rm 338 | Open |
| **Intro to MATLAB** | October 21, 10 AM - 12 PM | Sherbrooke 550 W, Rm 136 | Open |
| **Data Processing in Python (Part 2)** | October 28, 11 AM - 1 PM | Sherbrooke 550 W, Rm 136 | Open |
| **Intro to Machine Learning** | October 30, 1 - 3 PM | Sherbrooke 550 W, Rm 189 | Open |
| **More workshops to be announced soon!** | | | |

https://www.mcgill.ca/micm/training/workshops-series

# Outline

1. Data Wrangling
2. Linear Regression Models
3. Logistic Regression Models
4. Statistical Testing
5. Study Design

**Acknowledgements**

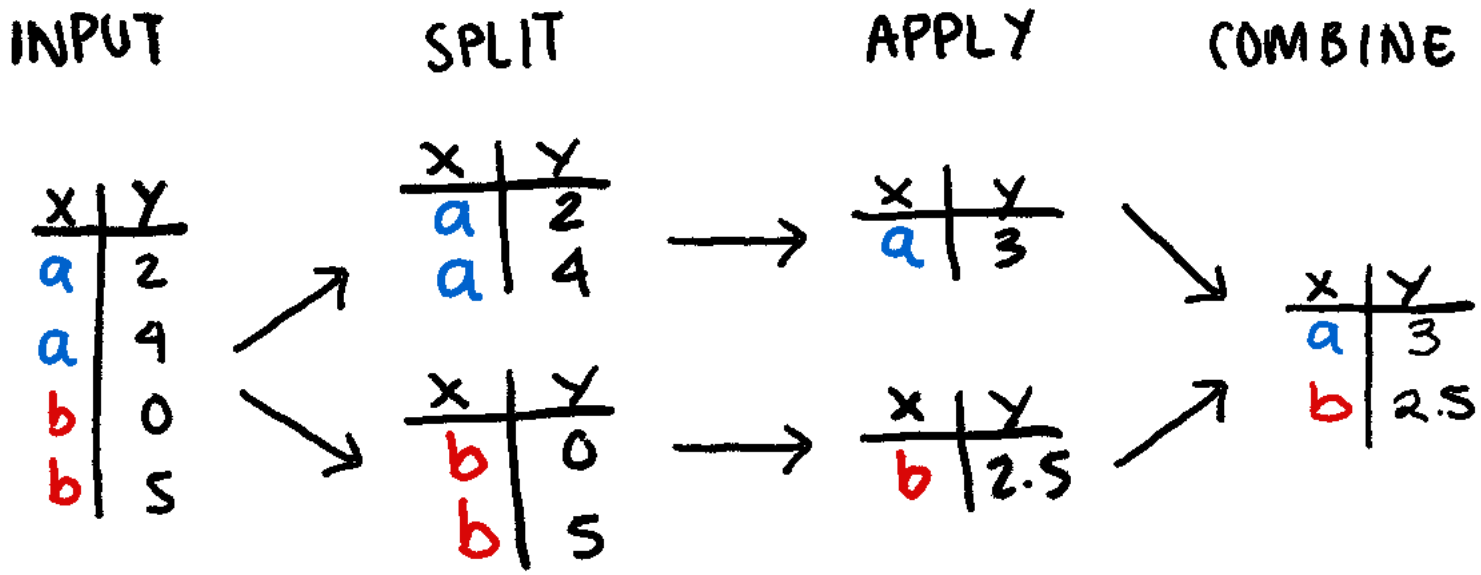
Gerardo Martinez - McGill
Alex Diaz-Papkovich - Brown
Larisa Morales Soto - HMS
Lisa Sullivan BUSPH

# Data Wrangling

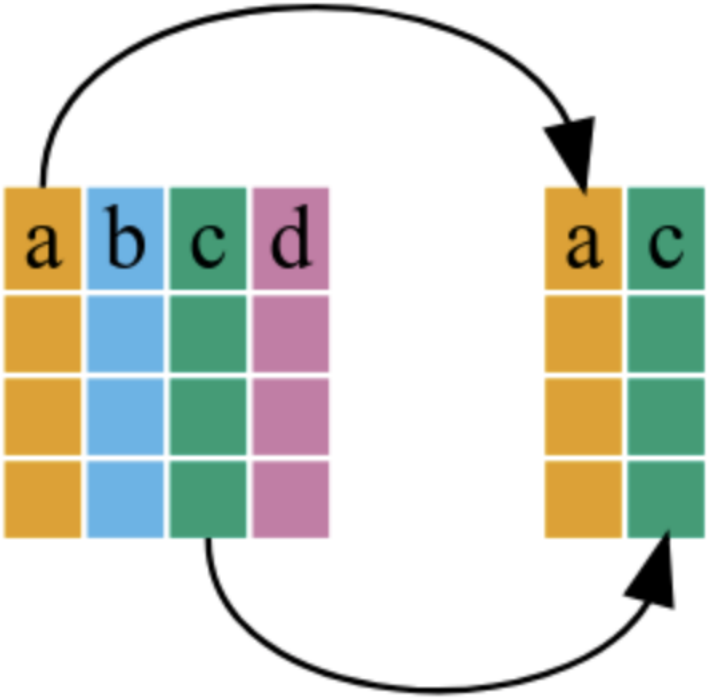
**Learning objectives:**

- Become familiar with the dplyr syntax
- Create pipes with the operator *%>%*
- Perform operations on data frames using dplyr and tidyr functions
- Examples for how to deal with missing data

# Split-Apply-Combine problem

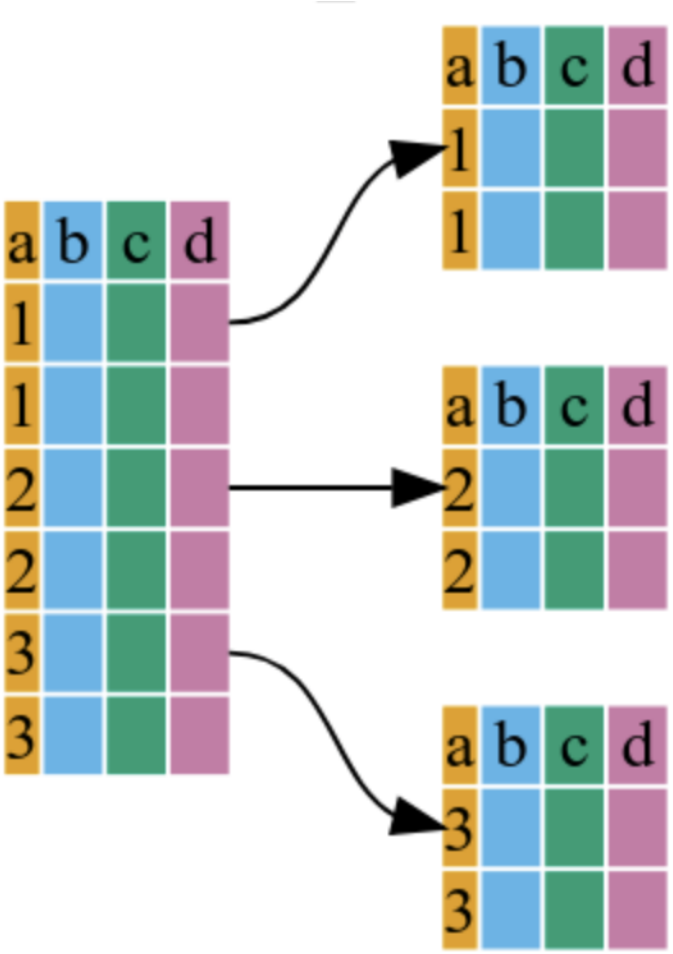


# Select



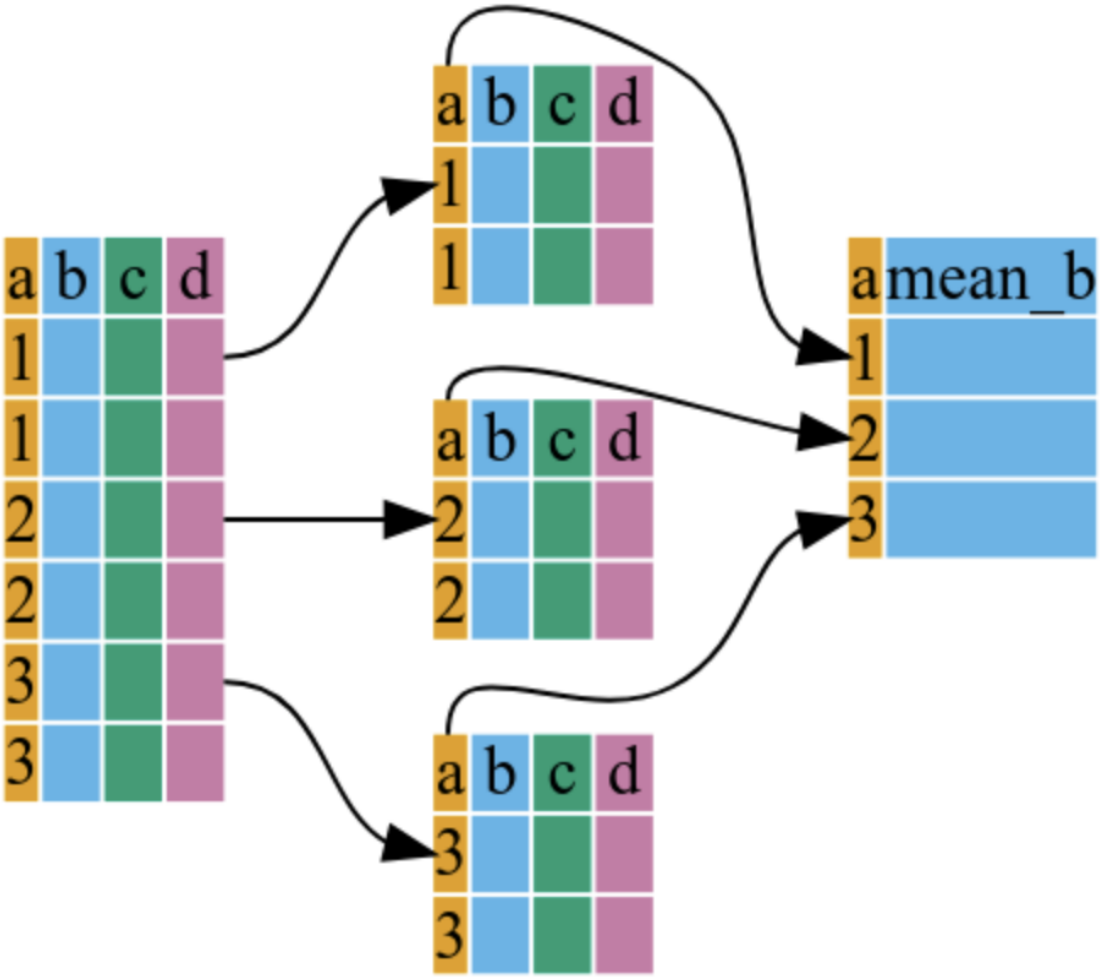
https://swcarpentry.github.io/r-novice-gapminder/13-dplyr/index.html

# Group by



https://swcarpentry.github.io/r-novice-gapminder/13-dplyr/index.html

# Summarize



# Managing Missing Data

- Most of the time, samples having missing entries
  - Missed medical appointment
  - Error with measuring instruments etc.
  - Represented as ***NA*** in R

Simple Solution

- Drop samples with missing observations

More rigorous solutions

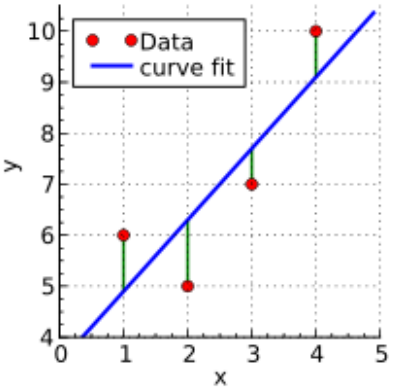
- Fill in with average/median of dataset or condition
- Train a model to 'impute' missing entries based on other features

# Linear Regression

**Learning objectives:**

- Understand the fundamental regression algorithms
- What is a coefficient/effect size

# The Classic Normal Linear Model



Credit: Wikipedia

- The fundamental model for statistics & machine learning
- Follow **a Normal distribution**

Imagine a dataset with a dependent variable ***y*** and a set of descriptive features ***x***

We want to learn what features in *x* are good predictors of *y* (what is their **effect size/coefficient** β)

$$y = \beta x + \epsilon \qquad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

$$\mathrm{E}(y|x) = \beta x \qquad y|x = \mathcal{N}(\beta x, \sigma^2)$$

# Model Assumptions

1. Outcome is **continuous** and a **linear combination of predictors**
2. Outcome is such that $y_i|x_i = \mathcal{N}(\beta x_i, \sigma^2)$
3. Predictors must not be perfectly correlated (linear combination) ----- avoid multicollinearity
4. For every observation *i,* the error is:
    1. **Normally distributed**
    2. **Mean zero**
    3. **Homoskedastic** (same variance as other observations)
    4. **Independent** (not correlated)

Why?

$$\hat{\beta} = (x^T x)^{-1} x^T y$$

If columns of $x$ are linearly dependent on each other, $x^T x$ is non-invertible!

# Model Fitting

Probabilistic Approach

- **Maximum Likelihood Estimation**

$$\ell(\beta, \sigma^2 \mid y, X) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}(y - X\beta)^\top(y - X\beta)$$

$$\hat{\beta} = \arg_\beta \ell$$

Machine Learning

- **Ordinary Least Square**

$$y = \beta x + \epsilon$$

$$\hat{y} = \hat{\beta} x$$

$$MSE = \frac{\sum_{i=1}^{N} (\hat{y}_i - y_i)^2}{N}$$

# Extensions

More than Normal distribution! However, if we lose the normality for error term (residual), we will fail to do hypothesis tests.

Interaction effects

- Outcomes depend on interactions of (e.g.) age*sex

Popular Models for Different Types of Data

- **Multivariate regression (multiple, correlated outcomes)**
- **Logistic regression (Binary outcomes like healthy vs. disease)**
- Multinomial/Poisson regression (Count data)
- ARIMA (Time series—data correlated over time)

# Logistic Regression

**Learning objectives:**

- What is a logistic function
- How LR extends the linear model
- What to do when we have more than 2 categories

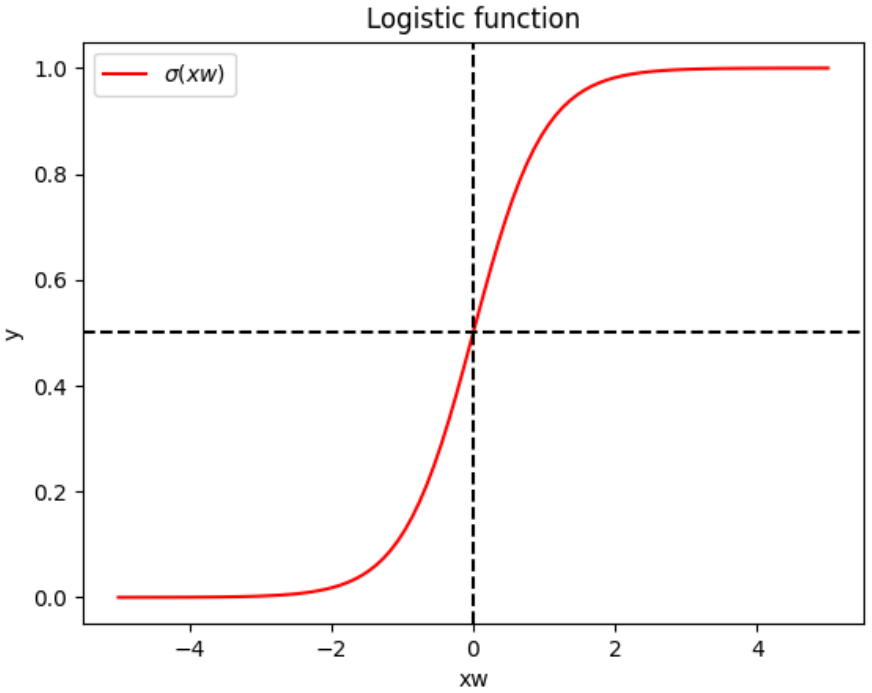
Linear Regression could be used to predict counts (technically) BUT it has many flaws

- Formulation isn't bounded (0,1)
- Impractical for predicting categories

If we have binary outcomes (0 or 1), logistic regression is a better option!

# Logistic Regression (Generalized Linear Model)

- Sigmoid/Logistic function provides us with a bounded output
- Range is also easy to interpret as a probability




Default link function:

$$y \sim \text{Bernoulli}(p)$$

$$\mu = p = P(y = 1|x)$$

$$\ln\left(\frac{p}{1-p}\right) = X\beta$$

$$P(Y = y \mid x) = p^y(1-p)^{1-y}, \quad y \in \{0, 1\}$$

# More than two categories

Poisson and negative binomial models are good options.

$$y \sim \text{Poisson}(\lambda)$$

$$\mu = \lambda = E(y \mid x)$$

$$\ln(\lambda) = X\beta$$

$$P(Y = y \mid x) = \frac{e^{-\lambda}\lambda^{y}}{y!}, \quad y \in \{0, 1, 2, \ldots\}$$

$$y \sim \text{NegativeBinomial}(\mu, r)$$

$$\mu = E(y \mid x)$$

$$\ln(\mu) = X\beta$$

$$P(Y = y \mid x) = \binom{y + r - 1}{y} \left(\frac{r}{r + \mu}\right)^{r} \left(\frac{\mu}{r + \mu}\right)^{y}, \quad y \in \{0, 1, 2, \ldots\}$$

where $r > 0$ is the **dispersion parameter** controlling overdispersion.

# Statistical Testing

**Learning objectives:**

- Understand the formulation an assumption of standard statistical tests
- Understand the interpretation of a p-value
- Considerations when designing an experiment

# Hypothesis Tests

**Goal**

- With a selected significant level, if two datasets show heterogeneity
- How to approach this by using **test statistics**

**Many Types of Tests**

- Parametric tests – **make assumptions on the data distribution**
  - Z-score vs. t-test
  - ANOVA (F-test)
- Non-parametric – **does not make assumptions on data dist.**
  - Permutation test

# Hypothesis Formulation

**Null Hypothesis**

- Effect size is 0
- Difference between conditions is 0

**Alternative Hypothesis**

- Effect size is NOT 0
- Difference between conditions is NOT 0

We **accept or reject** the null based on the generated **p-value (< 0.05 – this is self-selected)**

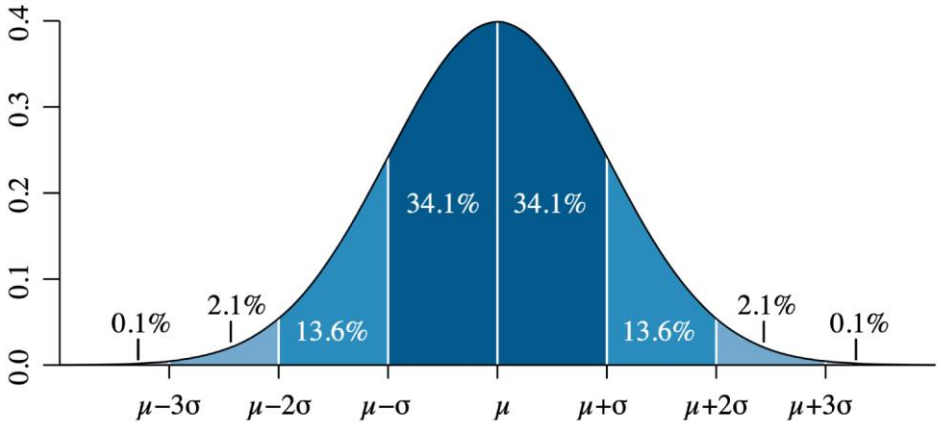
# Understanding Significance

Statistical significance **DOES NOT** imply causality

It is a measurement of the likelihood of the observed data happening by chance.

A p-value tells us the probability of seeing the observation **if the null hypothesis is true.**

# Z-score Test

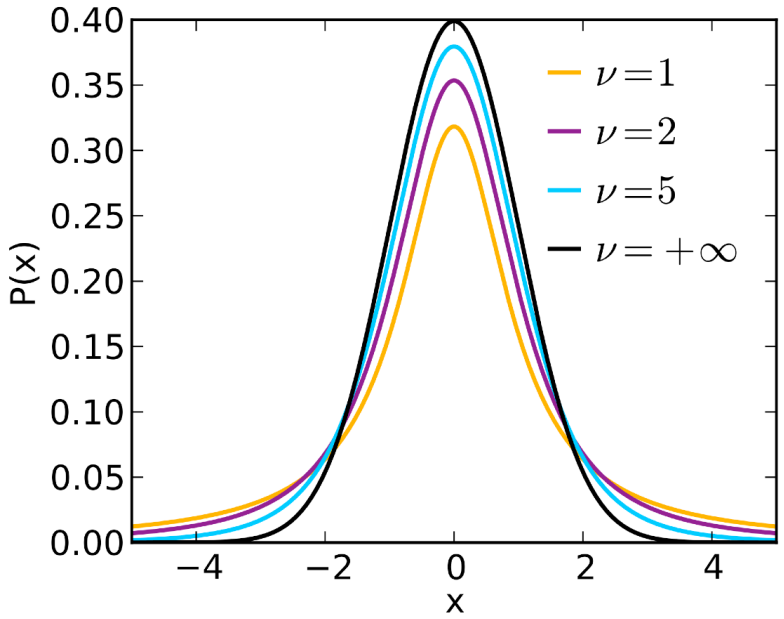



Credit: Wikipedia

- Based on the Normal Distribution
- Generate a Z-score from our effect size estimate
- Used when you know your data follows a normal distribution or data size is large so that you can approximate a Normal distribution (central limit theorem)
- Compare to Normal(0,1) distribution
  - One-tail or Two-tail

$$Z = \frac{\beta - \hat{\beta}}{\mathrm{SE}(\beta)}$$

# t-test




Credit: Wikipedia

Based on the student t distribution

- Defined by degrees of freedom (d.f)
  - n-1 d.f.
- More appropriate for smaller sample sizes (most cases)
- Use our sample mean $\overline{X}$ and sample s.d $\hat{\sigma}$ to calculate test statistic
- Value of $\mu$ depends on your null hypothesis
  - One-Sample - 0
  - Two-sample - $\mu_2$

$$t = \frac{\overline{X} - \mu}{s} = \frac{\overline{X} - \mu}{\hat{\sigma}/\sqrt{n}}$$

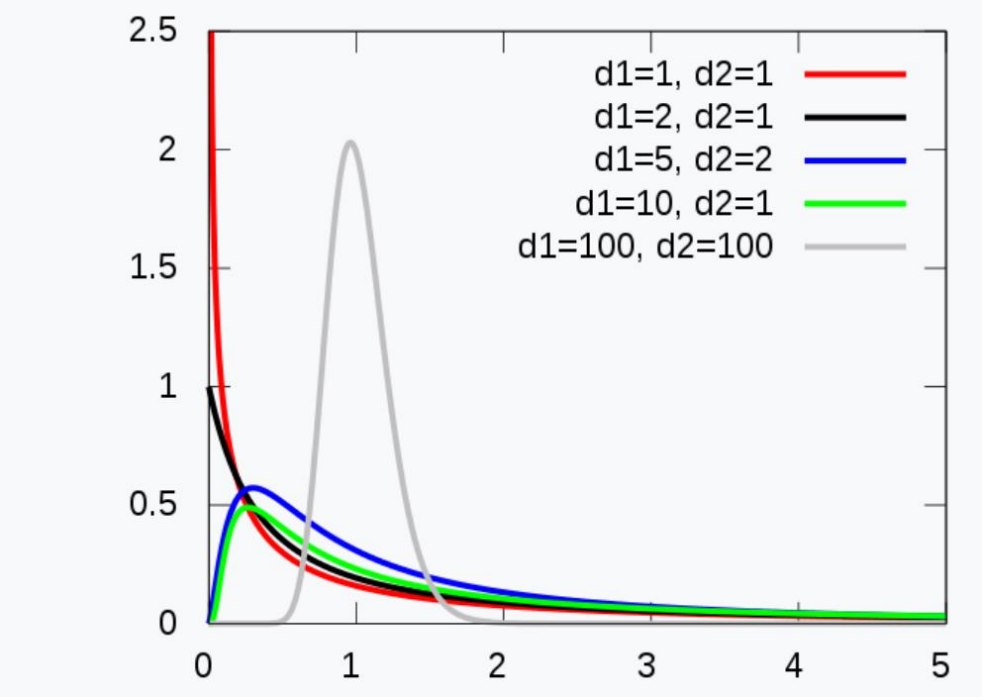
# ANOVA

Based on the F-distribution

- Similar to t-statistic
- Statistic is the ratio of two Chi-squareds

Null hypothesis: multiple means are **all** equal

Alternative: At least one group is different




Credit: Wikipedia

# ANOVA

How much variation is explained by our treatment condition?

$$SST = \sum_{i=1}^{n} (y_i - \overline{y})^2$$

Total sum of squares

$$SST = SS(Treatments) + SS(Residuals)$$

Take the ratio of the variance (mean sum of squares) between treatments over the variance within treatments

High variance between treatments will give a low p-value

# Non-Parametric Test

Similar to what is done for enrichment analyses

- Avoids needing to make assumptions on distributions
- Can be very **slow**

1. Shuffle labels in your data to generate estimates (random baseline)
2. See where your true observation lies on this baseline

# Multiple Testing

- May test multiple hypothesis (example, effect sizes in linear model)
- If null distribution is true, expect a **uniform distribution** of p-values
  - 5% of tests will give a significant result
- Multiple test correction methods are used to address this
  - Bonferonni Correction
  - **Benjamini & Hochberg Correction**
  - Found in results as **FDR, p-adjusted, q-value**

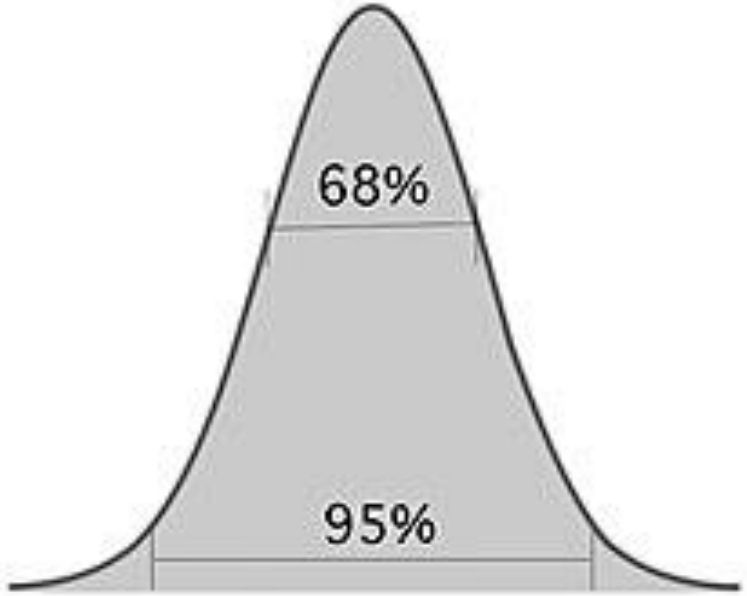
# Confidence Intervals

- We may want to attribute a range to our estimate

- Can be calculated with the **margin of error**

$$E = Z\frac{\sigma}{\sqrt{n}}$$

- Where Z is the z-score that defines our confidence level

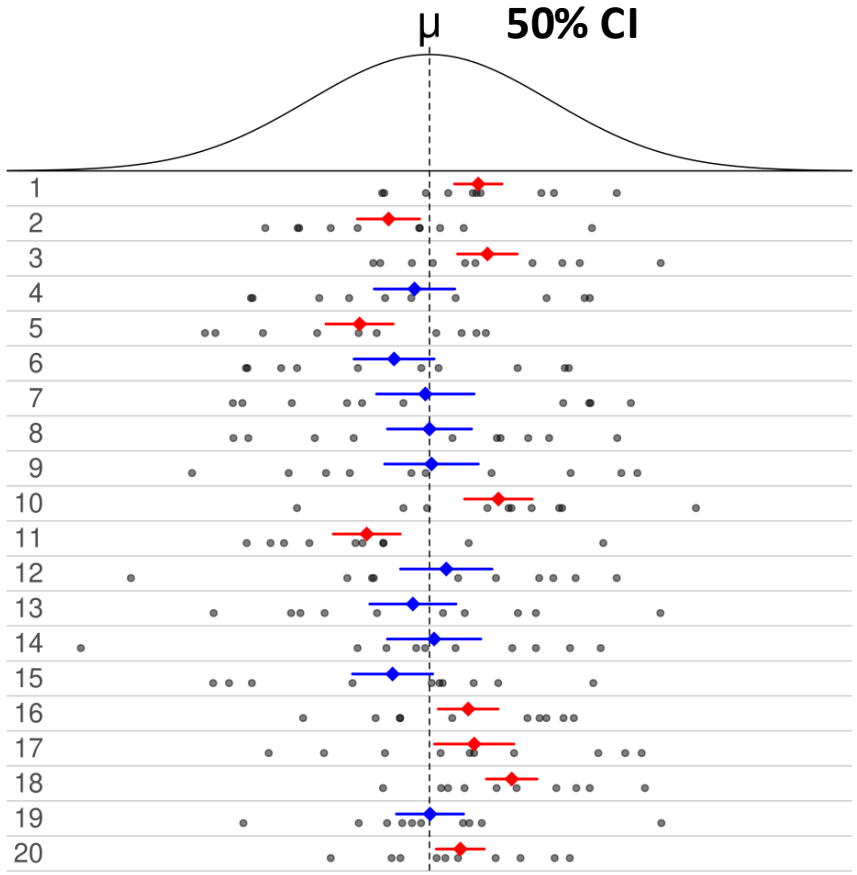


# PSA: Confidence Intervals

A 95% CI **DOES NOT mean** there is a 95% chance the true value is in that range

**For multiple experiments, 95% of CIs will overlap with the true value**

**Consider a Bayesian formulation instead (beyond scope)**



# Study Design

**Learning objectives:**

- Understand how simulations can be used to help with study design
- How to determine the required sample size

# Study Design

Data collection can be expensive and time-consuming

Need to ensure that the collected data has enough **power** to avoid false negatives (type II error)

Can simulate data to help decide on the statistical test used and determine the appropriate **sample size**

# Determining sample size

- Statistical test results depend on multiple factors
  - Difference in Samples
  - **Sample Size**
- We want to make sure we have enough samples to avoid False Negatives (type II errors)

- Can leverage the assumptions used for our statistical test to determine sample size

# Confidence Intervals

In practice you will want your CI to cover a small range to be informative.

Can achieve this by solving for the **margin of error E**

$$E = Z\frac{\sigma}{\sqrt{n}}$$

$$n = \left(\frac{Z\sigma}{E}\right)^2$$

Note that we may not have a value for σ

- Use estimate from past study
- Make an assumptive guess
- Run a small pilot study

# Sample Comparison

$$n_i = 2\left(\frac{Z_{1-\frac{\alpha}{2}} + Z_{1-\beta}}{ES}\right)^2$$

$$Effect\ Size\ (ES) = \frac{|\mu_1 - \mu_2|}{\sigma}$$

$\alpha \rightarrow$ *Significance Level*

*Probability of rejecting null when it is true*

$1\text{-}\beta \rightarrow$ *Power*

*Probability of rejecting null when it is false*

# To summarize

- ✓ Basics for data manipulation & cleaning
- ✓ Explored basic linear modeling for regression & classification
- ✓ Explored statistical testing and intuition behind study design

**Now you are ready to:**

- Use dplyr to facilitate reproducible data manipulation
- Apply and interpret linear model results
- Interpret confidence intervals and p-values
- Explore more complex models and tests!

# Future Statistics Workshops

- Dimension Reduction
- **Bayesian Inference**
- Time-series Analysis
- And more…

# Thank you for attending!

Scan the QR code to confirm you attended today's workshop.
Fill out the survey to help us improve our offerings!